SIEVE OF ERATOSTHENES

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

From the Sieve of Eratosthenes, we observe that,

- The crossed-out numbers are ***composite*** and the ***coloured numbers*** (encircled) are ***primes.*** The total number of primes upto 100 is 25.
- The only prime number that ends with 5 is 5.

1.2.2 Expressing a Number as the Sum of Prime Numbers

Any number greater than 3 can always be expressed as the sum of two or more prime numbers. Let us illustrate this in the following examples.

Example 1: Express 42 and 100 as the sum of two consecutive primes.

Solution: 42 = 19+23;

100 = 47+53

Example 2: Express 31 and 55 as the sum of any three odd primes.

Solution: 31 = 5+7+19 (find another way, if possible!)

55 = 3 + 23+29

Try these

(i) Express 68 and 128 as the sum of two consecutive primes.

(ii) Express 79 and 104 as the sum of any three primes.

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